

Two-Secret Image Sharing Scheme by Shifting Random Grids*

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Abstract

Visual secret sharing (VSS) was proposed by Naor and Shamir in 1995. Visual secret sharing schemes encode a secret image into two or more share images, and single share image can't obtain any information about the secret image. When superimposes the shares, it can restore the secret by human vision. Due to the traditional VSS have some problems like pixel expansion and the cost of sophisticated. The method of random grids was proposed by Kafri and Keren in 1987. But this method only can encode one secret image. Therefore, some schemes of encrypting more secret images into two shares were proposed. But those schemes are almost limited in decoding when those restored secret images have much distortion, like Chang et al.'s scheme in 2010. If there is too much distortion, we can't encrypt more information. In this paper, three new algorithms which based on Chang et al. are proposed. These algorithms can adjust distortion to infinitesimal. Also, they can distribute the distortion over two restored secret images. Moreover, the algorithms can achieve no distortion. These three algorithms not only retain the advantage of VSS but also improve on the problems of decoding.

1 Introduction

Recently, the development of network technology was very soon. In order to guarantee the secret image that transmits through the network will not be stolen, the secret images must be encrypted, the concept of this process is called secret image encryption. There are many research [9][11] about secret image encryption have been proposed. The studies of visual cryptography (VC) are one of method of secret image encryption proposed by Naor and Shamir [11] in 1995. VC can encrypt a secret image into two shares.

The secret information can be decrypted by directly stacking two share images without any computation in computer. However, these methods need to construct codebook and they have problem of pixel expansion. So, many researches began to use the random grid [9] algorithm to encrypt the secret image.

In 1987, Kafri and Keren proposed the encryption of a secret image by random grids. A binary secret image can turn into two meanings shares by random grids with same size of the original secret image. The decryption process is the same as that of VC. Therefore, more researches used by random grids were proposed. Chen et al. [4] (2008) proposed a RG-based multi-VSS which encrypts two secret images into two shares. However, Chen et al.'s scheme has the restriction on distortion when decrypting. So, Chang et al. [2] (2010) proposed a RG-based multi-VSS which not only encrypts two secret images as Chen et al.'s but also can adjust distortion with users. Unfortunately this scheme is still having too much distortion.

Hence, this paper proposes an improved multi-VSS scheme by shifting random grids. The scheme can adjust distortion to infinitesimal. And also can distribute the distortion over two restored secret images. Moreover, it can achieve no distortion. This scheme not only kept the advantage of the proposed scheme by Chang et al. but also improve on the problems of decoding.

2 Related Work

First of all, TABLE 1 defines all results when overlapping any two pixels, where 1 represent black, 0 represent white pixels.

TABLE I
ALL RESULTS WHEN OVERLAP TWO DIFFERENT PIXELS

r_1	r_2	$r_1 \oplus r_2$
0	0	0
0	1	1
1	0	1
1	1	1

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In 1987, Kafri and Keren define the random grid of each pixel is only classified into transparent (white) or opaque (black) [9]. And each pixel is generated by the random number, so the amount of the transparent and opaque pixels will equal. Therefore the average light transmission ($\mathfrak{T}$) of a random grid is $1/2$. In which light transmission is defined as the percentage of incident light that passes through an image, that is the number of white pixels over all pixels of an image. Kafri et al. propose three different algorithms to encrypt a binary secret image. The input of the algorithm is the image B which want to encrypt, the output are two random grids for the R_1 and R_2 . The following are these three algorithms in [9]:

Algorithm 1:

Generate a $w \times h$ random grid R_1 // $\mathfrak{T}(R_1) = 1/2$
 for ($i = 0; i < w; i++$)
 for ($j = 0; j < h; j++$)
 if ($B[i][j] == 0$)
 $R_2[i][j] = R_1[i][j];$
 else $R_2[i][j] = R_1[i][j];$
 output (R_1, R_2)

Algorithm 2:

Generate a $w \times h$ random grid R_1 // $\mathfrak{T}(R_1) = 1/2$
 for ($i = 0; i < w; i++$)
 for ($j = 0; j < h; j++$)
 if ($B[i][j] == 0$)
 $R_2[i][j] = R_1[i][j];$
 else $R_2[i][j] = \text{random}(0, 1);$
 output (R_1, R_2)

Algorithm 3:

Generate a $w \times h$ random grid R_1 // $\mathfrak{T}(R_1) = 1/2$
 for ($i = 0; i < w; i++$)
 for ($j = 0; j < h; j++$)
 if ($B[i][j] == 0$)
 $R_2[i][j] = \text{random}(0, 1);$
 else $R_2[i][j] = R_1[i][j];$
 output (R_1, R_2)

Next, this paper discusses the staggering encryption Algorithms 4 by moving horizontally. In order to explain clearly, this paper gives an example by using the Algorithm 1 to carry out the encryption in the proposed scheme. The inputs of our scheme are two images S_A, S_B which want to be encrypted, and the output of this scheme are two encrypted images G_1 and G_2 .

Subsequently, some definitions in Chang et al.'s Algorithm 4 [2] will be explained as follows.

Definition 1: $f_{RSP}(\cdot): Y \leftarrow f_{RSP}(X)$, Y is the output of

the function $f_{RSP}(\cdot)$ with the inputs X , where $f_{RSP}(\cdot)$ is that randomly select a pixel of X .

Definition 2: $f_{RG}(\cdot): Y||Z \leftarrow f_{RG}(X)$, Y and Z are the outputs of the function $f_{RG}(\cdot)$ with the input X , where $f_{RG}(\cdot)$ is one of the three random grids algorithm in [8] which inputs a pixel of the secret image, then outputs two cipher-pixels for two shares.

Definition 3: $\bar{f}_{RG}(\cdot): Z \leftarrow \bar{f}_{RG}(X, Y)$, Z is the output of the function $\bar{f}_{RG}(\cdot)$ with the inputs X and Y , where $\bar{f}_{RG}(\cdot)$ is the function according to $f_{RG}(\cdot)$ (as in Definition 2) which inputs a cipher-pixel of one share Y and a pixel of the secret image X , then outputs the other cipher-pixel.

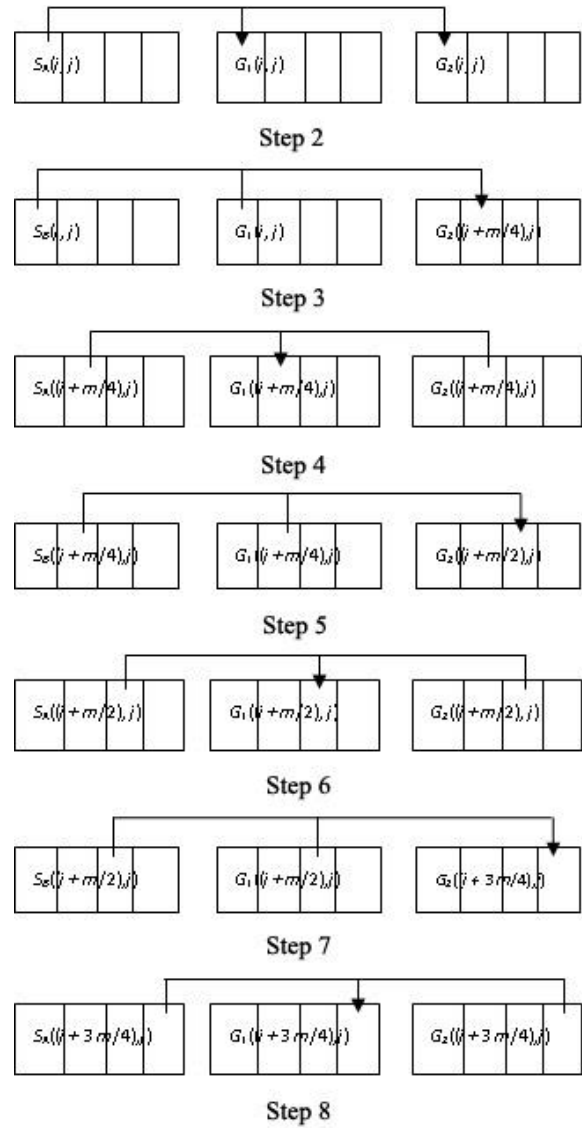


Fig. 1 The diagram of the processes in the encryption phase of the proposed scheme with respected to moving horizontally $1/4$ width.

As stated above, Chang et al. propose an Algorithm 4 that encrypts two images simultaneously to obtain two shares of confidential images, when superimpose directly can obtain the first secret image. The second segment will move horizontally and then superimpose the first fragment, will receive the second secret image. The following is the method that the second share moves horizontally $1/4$ width when restored the second image. The idea of their scheme is presented in Figure 1.

Algorithm 4:

Step 1: $S_A(i, j) \leftarrow f_{RSP}(S_A)$.
Step 2: $G_1(i, j) \| G_2(i, j) \leftarrow f_{RG}(S_A(i, j))$.
Step 3: $G_2((i + m/4), j) \leftarrow \bar{f}_{RG}(S_B(i, j), G_1(i, j))$.
Step 4: $G_1((i + m/4), j) \leftarrow \bar{f}_{RG}(S_A((i + m/4), j), G_2((i + m/4), j))$.
Step 5: $G_2((i + m/2), j) \leftarrow \bar{f}_{RG}(S_B((i + m/4), j), G_1((i + m/4), j))$.
Step 6: $G_1((i + m/2), j) \leftarrow \bar{f}_{RG}(S_A((i + m/2), j), G_2((i + m/2), j))$.
Step 7: $G_2((i + 3m/4), j) \leftarrow \bar{f}_{RG}(S_B((i + m/2), j), G_1((i + m/2), j))$.
Step 8: $G_1((i + 3m/4), j) \leftarrow \bar{f}_{RG}(S_A((i + 3m/4), j), G_2((i + 3m/4), j))$.
Step 9: Repeating **Step 1 to 8** until all the cipher-pixels of two cipher-grids are generated.

3 Main Result

This paper proposes three algorithms which are based on Algorithm 4 also can encrypt two secret images into two shares. The Algorithm 5 can reduce the distortion to $1/p$. And Algorithm 6 let distortion distribute over two restored secret images. Algorithm 7 does even achieve no distortion.

A. Encryption phase

According to the proposed method, the distortion $1/4$ can be reduced to smaller. The distortion of the second restored image can be chose by users. The amount of distortion is related to the degree of moving when we restored the second image. For example, if the distortion was $1/4$, we can set the degree of moving to be quarter, half or three quarters of the width of the second share. For the same reason, if we want to reduce the distortion to $1/10$, we can set the degree of moving to be p tenth, for $p = 1, 2, \dots, 9$. This way can get more clearly restored secrete

image.

The inputs of the Algorithm 5 are two images S_A, S_B with the size of $m \times m$ be encrypted into two share images G_1 and G_2 with the size of $m \times m$ without any pixel expansion. And the input p is the degree of we want to move. That is to say, we can reduce the distortion to $1/p$. The details of Algorithm 5 are as following:

Algorithm 5($S_n = S_A, p$):

$S_A(i, j) \leftarrow f_{RSP}(S_A)$
 $G_1(i, j) \| G_2(i, j) \leftarrow f_{RG}(S_A(i, j))$
 int a = 0, b = 1;
 for(int i = 3; i <= p * 2; i++)
 if(i % 2 == 1)
 $G_2((i + m*b/p), j) \leftarrow \bar{f}_{RG}(S_B((i + m*a/p), j), G_1((i + m*a/p), j))$
 else
 $G_1((i + m*b/p), j) \leftarrow \bar{f}_{RG}(S_A((i + m*b/p), j), G_2((i + m*b/p), j))$
 a++;
 b++;

Repeat Algorithm 5 until all the cipher-pixels of two cipher-grids are generated.

Next, Algorithm 6 make that distortion be distributed over two restored secret images as following:

Algorithm 6:

Step 1: Randomly select one of the secret images $S_n(i, j)$, where $n = A$ or B . All steps of the S_A can substitute for S_B , and vice versa.

Step 2: Call Algorithm 5(S_n, p)

Step 3: Repeat Algorithm 6 until all the cipher-pixels of two cipher-grids are generated.

Algorithm 7 does even achieve no distortion as following:

Algorithm 7:

for (i = 0; i < w; i++)
 for (j = 0; j < h; j++)
 $G_1(i, j) \| G_2(i, j) \leftarrow f_{RG}(S_A(i, j))$
 int a = 0, b = 1;
 for(int i = 3; i <= p * 2 - 1; i++)
 if(i % 2 == 1)

$$G_2((i + m*b/p), j) \leftarrow \bar{f}_{RG}(S_B((i + m*a/p), j), G_1((i + m*a/p), j))\}$$

else

$$G_1((i + m*b/p), j) \leftarrow \bar{f}_{RG}(S_A((i + m*b/p), j), G_2((i + m*b/p), j))\}$$

a ++;

b ++;

A. Decryption phase

Upon collecting these two cipher-grids G_1 and G_2 , the users can easily restore the first secret image S_A by directly superposing G_1 and G_2 . The second secret image S_B can be restored by superposing G_1 and G_3 , where G_3 is obtained from G_1 by moving horizontally m/p width.

This paper proposes a method of RG-based multi-VSS which encrypts two secret images into two shares. In the following section, we present some results to show that the distortion of the restored images can be reduced as one want, which is better than Chang et al.'s scheme [2].

4 Experiments

Simulation 1:

In this simulation, we take $p = 10$ for example as following:

Two secret images S_A and S_B with the size of 400×300 are stored some simple English alphabets, Figure 2(a) and 2(b), which are encrypted into two cipher-grids shares G_1 and G_2 with the size of 400×300 as shown in Figure 2(c) and 2(d).

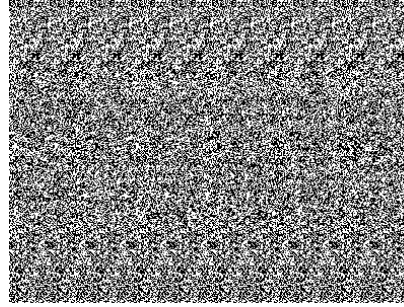
The first secret image S_A with the size of 400×300 is decrypted by superposing G_1 with G_2 directly as shown in Figure 2(e). Upon moving $1/4$ width for G_2 horizontally, the second secret image S_B with the size of 400×300 is decrypted by superposing G_1 and the moved G_2 , as shown in Figure 2(f).

CSIE

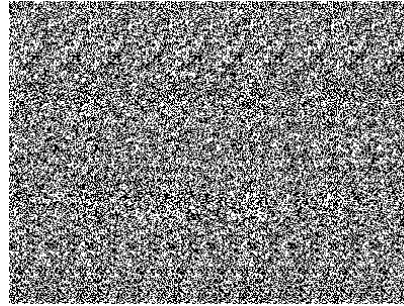
(a)

NCNU

(b)



(c)



(d)

CSIE

(e)

NCNU

(f)

Fig. 2 Results of Simulation 1: (a) the first secret image S_A , (b) the second secret image S_B , (c) share G_1 , (d) share G_2 , (e) the first restored secret S_A , (f) the second restored secret S_B .

Simulation 2:

In this simulation, we take $p = 50$. That is, moving horizontally $1/50$ width and the distortion can be spread for all restored images.

Simulation 3:

In this simulation, we also take $p = 50$. But this time we can see that there is no distortion since we made the secret in the middle of the secret image.

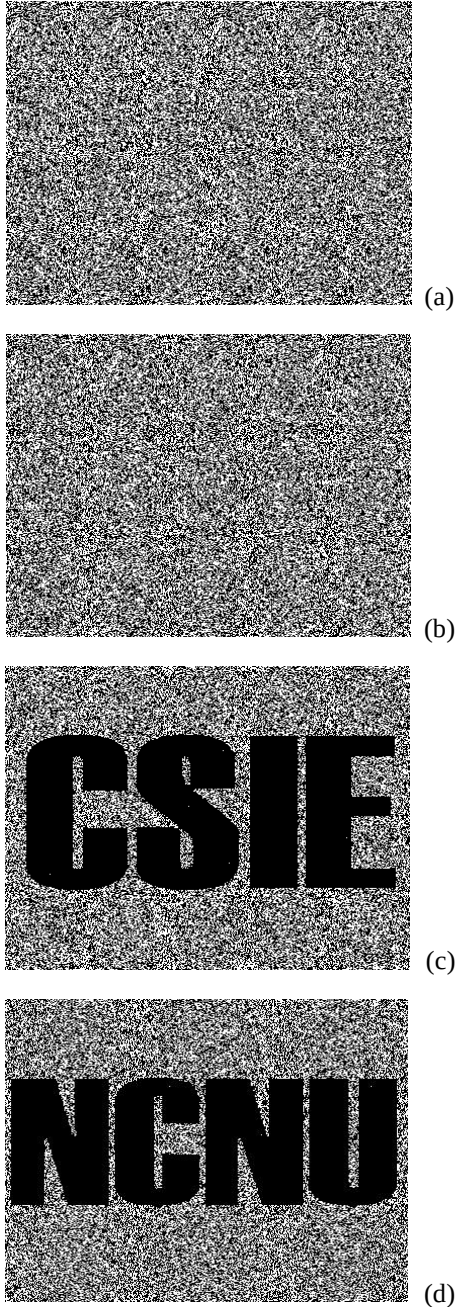


Fig. 3 Results of *Simulation 2*: the first secret image S_A is the same as *Simulation 1*, the second secret image S_B is the same as *Simulation 1*, (a) share G_1 , (b) share G_2 , (c) the first restored secret S_A , (d) the second restored secret S_B .

Likewise, *Simulation 2* and *3* show the cases of

moving horizontally by $1/50$ width when restore the second secret image. The experimental results are given in Figure 3 and 4, respectively.

5 Conclusion

In this paper, a novel visual secret sharing (VSS) scheme with high capability of encryption is proposed. The proposed scheme encrypts two secret images at once into two shares. When restored the secret image, first one can be got by overlap these two shares directly, while the second one can be got by overlap the first share with the shifted second share, where shift means moving horizontally.

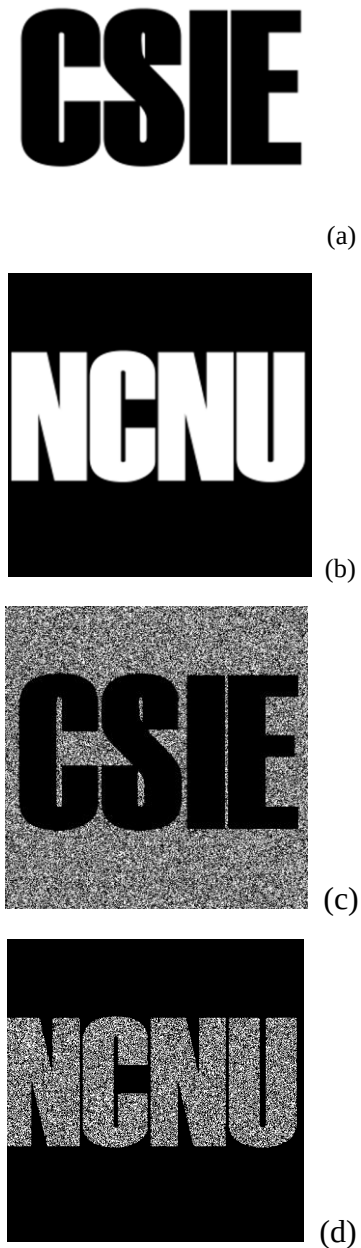


Fig. 4 Results of *Simulation 3*: (a) the first secret image S_A , (b) the first restored secret S_A , (c) the second secret image S_B , (d) the second restored secret S_B .

Compared with the traditional VSS methods such as VC and random grids, the proposed scheme needs not to redesign any codebook and without any pixel expansion. Moreover, the proposed scheme can carry more information than traditional VSS schemes and saves the cost of transmission bandwidth and storage.

Compared with the VSS methods which are proposed by Chang-Chou Lin and Wen-Hsiang Tsai [8] in 2002. This scheme used image processing to transform gray-level images into an approximate binary image, but when encrypting the secret images, it has a problem which is pixel expansion.

In this paper, the simulate results of superposing these two shares are displaying on the computer. These algorithms can adjust distortion to infinitesimal. Also, they can distribute the distortion over two restored secret images. Moreover, the algorithms can achieve no distortion. These three algorithms not only retain the advantage of VSS but also improve on the problems of decoding.

In the near future, we will try to encrypt limitless secret images into two shares based on the proposed Algorithm 5 to Algorithm 7. Hence, the proposed scheme can be widely applied.

References

- [1] J.-L. Bai, "Random-based secret image sharing scheme," *Master's Thesis*, Computer Science and Information Engineering, Ming Chuan University, 2005.
- [2] Joy Jo-Yi Chang, Ming-Jheng Li, Yi-Chun Wang, Justie Su-Tzu Juan*, Oct 2010, "Two-Image Encryption by Random Grids," *Prof. of 10th International Symposium on Communications and Information Technologies (ISCIT2010)*, Meiji University, Tokyo, Japan, Oct. 26-29, 2010, pp. 458-463. (NSC98-2221-E-260-013-) (EI)
- [3] T.-H. Chen, G.-Z. Wei, and K.-X. Taso, "An multi-secret image scheme by using random grids," in *Proceedings of 18th Information Security Conference*, Hualien, May 29-30, 2008.
- [4] T.-H. Chen, K.-H. Tsao, and K.-C. Wei, "Multiple-image encryption by rotating random grids," in *Proceedings of The 8th International Conference on Intelligent System Design and Applications (ISDA 2008)*, vol. 3, 2008, pp. 252-256.
- [5] T.-H. Chen, and K.-H. Tsao, "Image encryption by (n , n) Random Grids," in *Proceedings of 18th Information Security Conference*, Hualien, May 29-30, 2008.
- [6] T.-H. Chen, K.-H. Taso, and Y.-T. Yang, "Friendly color visual secret sharing by random grids," *Fundamenta Informaticae*, vol. 96, 2009, pp. 61-70.
- [7] Y.-Y. Chen, Justie S.-T. Juan, and M.-J. Li, "A random grid-based visual cryptography to adjust light transmission and the number of shares," in *Proceeding of 2009 National Computer Symposium*, National Taipei University, Taiwan, Nov. 27-28, 2009, pp. 184-192.
- [8] Lin, C. C. and Tsai, W. H. "Visual cryptography for gray-level images by dithering techniques," *Pattern Recognition Letters*, Vol. 24, pp.349-358, 2003.
- [9] O. Kafri, and E. Keren, "Encryption of pictures and shapes by random grids," *Optics Letters*, vol. 12, no. 6, 1987, pp. 377-379.
- [10] R. Lukac, and K. N. Plataniotis, "Bit-level based secret sharing for image encryption," *Pattern Recognition*, vol. 38, no. 5, 2005, pp. 767-772.
- [11] M. Naor, and A. Shamir, "Visual cryptography," in *Proceedings of Advances in Cryptology – Eurocrypt'94*, Lecture Notes in Computer Science, vol. 950, Springer Berlin, 1995, pp. 1-12.
- [12] S.-J. Shyu, "Image encryption by random grids," *Pattern Recognition*, vol. 40, no. 3, 2007, pp. 1014-1031.
- [13] S.-J. Shyu, "Image encryption by multiple random grids," *Pattern Recognition*, vol. 42, no. 7, 2009, pp. 1582-1596.